REVIEWER PIIP

PCIS2301 PROFESSIONAL ISSUES IN INFORMATION SYSTEMS
SIMPLIFIED REVIEWER IN BULLETED FORM

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MODULE 1: LAW AND THE COMPUTING PROFESSION
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1.0 LAW AND GOVERNMENT

- Law is a formal system of rules established by a governing authority
- Maintains social order, protects rights, resolves conflicts peacefully
- For IT professionals, law sets boundaries for what can be built, how data is handled, and liability for system failures
- Prevents tech monopolies and rogue developers from operating without accountability

1.1 SOURCES OF LAW

- Statutory Law
  - Formal written laws enacted by legislative bodies like Parliament or Congress
  - Example: Computer Misuse Act 1990, RA 10173, RA 10175
  - Explicitly written and passed by legislatures

- Common Law
  - Laws developed by judges through court decisions and precedent
  - Technology evolves faster than parliaments can write statutes
  - IT legal disputes often settled by judicial precedent
  - If a judge rules a specific action is illegal, that ruling becomes common law for future cases

- Administrative or Regulatory Law
  - Laws and rules created by government agencies
  - Agencies authorized by statutory laws to implement specific policy mandates
  - Example: Information Commissioner's Office defines what "secure" means under data protection laws

1.2 THE LEGISLATIVE PIPELINE

- Proposal
  - Inception phase where idea, demand, or policy prompts a White Paper
  - White Paper outlines core intent and scope before formal legislation

- First Reading
  - Drafted law (now called a Bill) is formally introduced
  - No debate, purely a public announcement

- Second Reading
  - First major debate phase
  - Lawmakers discuss principles, societal impact, and economic feasibility

- Committee and Report Stage
  - "Debugging" phase of the legislative pipeline
  - Rigorous line-by-line review
  - Committees call external experts like IT professionals for testimony
  - Experts provide technical fixes and eliminate loopholes
  - Technology law not built by politicians alone
  - Technical expertise is highly sought after here
  - Governments rely on tech professionals to avoid unfeasible or impossible laws

1.3 REGULATORY BODIES

- Parliaments write laws but lack technical expertise for daily enforcement
- Regulatory bodies bridge legislation and corporate execution
- Independent or quasi-governmental agencies overseeing specific industries
- Enforce administrative law
- In IT: statute might say "secure user data," regulatory body defines what "secure" means
- Example: mandating AES-256 encryption and auditing companies

International Tech Regulators
- Information Commissioners Office (UK) - data protection and privacy
- Federal Trade Commission (US) - consumer protection, deceptive cybersecurity practices
- Professional Institutes (BCS, ACM) - Royal Charters, self-regulatory bodies, professional conduct, education standards, membership grades

1.4 PRIMARY PHILIPPINES IT LAWS

- Data Privacy Act of 2012 (RA 10173)
  - Protects privacy while ensuring free flow of information
  - Organizations must implement strict security measures

- Must obtain explicit consent from data subjects

- Cybercrime Prevention Act of 2012 (RA 10175)
  - Criminalizes computer crimes
  - Hacking, identity theft, data interference, online libel
  - Establishes Cybercrime Investigation and Coordinating Center

- Electronic Commerce Act of 2000 (RA 8792)
  - Legal recognition of electronic documents, signatures, and transactions
  - Same legal weight as paper-based forms

- DICT Act of 2015 (RA 10844)
  - Established Department of Information and Communications Technology
  - Primary agency for ICT policy, planning, and development

1.5 SECTOR SPECIFIC REGULATIONS

- Financial Sector
  - Bangko Sentral ng Pilipinas issues circulars (808 and 982)
  - Mandate IT risk management and cybersecurity standards for banks and fintech

- Health Sector
  - Department of Health and PhilHealth guidelines
  - Govern electronic medical records in compliance with Data Privacy Act

- Intellectual Property
  - Intellectual Property Code (RA 8293)
  - Protects software copyrights and digital content

1.6 GOVERNANCE AND COMPLIANCE FRAMEWORKS

- National Privacy Commission 5 Pillars of Compliance
  - Appointing a Data Protection Officer
  - Conducting Privacy Impact Assessments
  - Creating a Privacy Manual
  - Implementing security measures
  - Establishing a breach management protocol

- National Cybersecurity Plan
  - Strategic roadmap by DICT
  - Protects critical infrastructures and government networks

- Why frameworks exist
  - IS professionals manage intangible but high-value assets (data, algorithms, digital identities)
  - Assets are vulnerable to breaches, manipulation, misuse
  - Assets are powerful and can influence decisions, behavior, and rights
  - Understanding frameworks is not optional - it is core professional competency

1.7 GOVERNANCE VERSUS COMPLIANCE

- Corporate Governance
  - Internal framework of rules, relationships, systems, and processes
  - Created by Board of Directors, IT leadership, professional conduct frameworks
  - Proactive: driven by ethics, values, risk management, public trust
  - Example: internal ethics board reviewing AI bias before launch
  - Key question: "What should we do to be ethical professionals?"

- Legal Compliance
  - Adhering strictly to external laws, regulations, and industry standards
  - Dictated by statutory law, regulatory bodies, international frameworks
  - Reactive/Mandatory: driven by fines, prosecution, civil suits
  - Example: cookie consent banner to satisfy data protection rules
  - Key question: "What must we do to avoid going to court?"

- Key Takeaway
  - Governance is how organization chooses to manage itself based on profession and codes
  - Compliance is the baseline legal requirement
  - Cannot have good compliance without strong governance
  - Operating only for compliance means ignoring broader ethical duties

2.0 CRIMINAL LAW VERSUS CIVIL LAW

- Criminal Law
  - Focuses on offenses against society, state, or public order
  - IT examples: ransomware, malicious code, deliberate financial fraud
  - Initiated by State/Government prosecutors
  - Standard of proof: Beyond a reasonable doubt

  - Outcome: Punishment (imprisonment, criminal fines)

- Civil Law
  - Focuses on disputes between private parties
  - IT examples: breach of contract for buggy software, copyright infringement, leaked code
  - Initiated by injured private individual or corporation
  - Standard of proof: On the balance of probabilities
  - Outcome: Compensation (financial damages, injunctions)

- Key Takeaway
  - Single software failure or security breach can trigger both types of liability

2.1 THE COMPUTING PROFESSION

- Job
  - Activity or piece of work in exchange for financial compensation
  - Execution-oriented and transactional
  - Trading time, labor, or technical skills for a paycheck
  - Governed by corporate management and employment law
  - Anyone can hold one

- Profession
  - Specialized knowledge, rigorous educational standards
  - Commitment to public service over mere profit
  - Self-governing, Royal Charters in some jurisdictions
  - Professional bodies police their own members
  - Violating ethics can result in loss of professional status and title

- Royal Charter
  - Formal document issued by a monarch granting special status
  - Highest honor an organization can receive
  - Elevates body from "association" to prestigious institution
  - Three major effects:
    - Public Trust and Prestige - operates in public interest
    - Power to Charter Individuals - award CITP, CEng titles
    - Independent Legal Status - hold property, sign contracts, sue/be sued

- Four Functions of Professional Bodies
  - Governance and Ethics: create ethical rules, investigate and discipline members
  - Gatekeeping and Quality Control: set qualifications, experience requirements
  - Education and Knowledge Sharing: ensure relevant teaching, keep members updated
  - Public and Legal Advocacy: provide expert advice to lawmakers

2.2 BCS CODE OF CONDUCT

- Binding set of ethical and professional standards
- All members must adhere regardless of role, rank, or location
- Central mission: technology is "made good for society"
- Four core pillars:

- Pillar 1: The Public Interest
  - Primary responsibility to safety, well-being, and rights of public
  - Even over employer's desires
  - Protect public health, privacy, security, environment
  - Respect human rights and avoid discrimination
  - Promote equal access to IT benefits and digital inclusion
  - Example: raise alarm if AI system is biased against older applicants

- Pillar 2: Professional Competence and Integrity
  - Personal honesty and self-awareness of capabilities
  - Only accept work you are qualified for
  - Do not misrepresent skills or knowledge
  - Commit to CPD to keep up with evolving technology
  - Understand and comply with relevant legislation
  - Reject bribery and unethical inducements

- Pillar 3: Duty to the Relevant Authority
  - Fiduciary responsibility to employer or client
  - Carry out work with due care and professional judgment
  - Avoid and disclose conflicts of interest
  - Maintain absolute confidentiality
  - Warn in writing if professional judgment is overruled
  - Example: document and warn about launching app with known vulnerability

- Pillar 4: Duty to the Profession
  - Maintain reputation of IT as respected career

- Support colleagues in the industry
- Act with integrity and respect toward peers
- Advance public knowledge of technology
- Combat false or misleading tech statements
- Example: address code plagiarism rather than looking away

2.3 ACM CODE OF ETHICS AND PROFESSIONAL CONDUCT

- Global conscience and ethical baseline for computing profession
- 24 imperatives divided into four sections
- Not a checklist, but a dynamic framework for responsible choices

- Section 1: General Ethical Principles
  - Minimize negative consequences (cyberbullying, poverty, digital exclusion)
  - Mitigate risks of injury, data loss, system destruction
  - Be transparent about limitations, bugs, conflicts of interest
  - Prevent algorithms from perpetuating systemic bias

- Section 2: Professional Responsibilities
  - Systematically test software for high performance and security
  - Only accept tasks you are qualified for
  - Commit to ongoing learning
  - Honor user confidentiality and data ownership laws

- Section 3: Professional Leadership Mandates
  - Corporate profits must not overshadow societal safety
  - Foster culture where engineers can whistleblow
  - Point out ethical flaws in projects

- Section 4: Compliance with the Code
  - Members hold themselves and peers accountable
  - Willful violations can result in revoked credentials and membership

- Key Takeaway
  - Company can be 100% legally compliant while creating manipulative algorithms
  - ACM Code demands rising above compliance
  - Ask not just "Is this legal?" but "Is this right?"

2.4 CONTINUING PROFESSIONAL DEVELOPMENT (CPD)

- Ongoing cycle of reflection, target-setting, and execution
- Strict ethical and structural mandate, not optional career booster
- Knowing what you are doing today does not mean you will know in five years

- Nine Models of CPD (Kennedy's Framework)

  - Transmission (Low Autonomy, Top-Down)
    - Deficit: targets weaknesses, tied to managerial appraisal
    - Training: acquiring predefined technical skills from expert
    - Standard-based: demonstrating competence against rigid standards
    - Cascade: expert trains others, dilutes collaborative elements

  - Transitional (Individualized)
    - Award-bearing: formal higher education qualifications
    - Coaching/Mentoring: one-on-one development, builds professional trust

  - Transformative (High Autonomy, Collaborative)
    - Community of Practice: peer-based environments, shared experiences
    - Action Research: systematic inquiry into workplace practices
    - Emancipatory: complete ownership over learning pathway, considers social and moral impacts

- Three Analytical Perspectives (Bott and Taylor)
  - Power Relationships: employer mandates vs. professional ownership
  - Forms of Knowledge: transmitting existing vs. generating new knowledge
  - Purpose and Outcomes: accountability vs. genuine growth

- Computing Profession is Uniquely Volatile
  - Laws of physics and anatomy change slowly
  - Software frameworks, security threats, hardware shift completely every few years
  - Degree from a decade ago is structurally obsolete if not maintained
  - Outdated knowledge risks catastrophic failure

- Four Types of CPD Activities
  - Formal Learning: training courses, postgraduate qualifications, certifications
  - Informal Learning/Self-Study: reading journals, publications, tech reports

- Work-Based Learning: new responsibilities, mentoring, system overhauls
- Professional Activity: conferences, branch meetings, standard-setting bodies

- Enforceable Rule under BCS Code (Pillar 2)
- Neglecting CPD is a disciplinable offense
- Cannot claim ignorance if using outdated methods leads to breach
- Can be stripped of "Chartered" status

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MODULE 2: ORGANIZATIONAL STRUCTURE AND MANAGEMENT
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1.0 ROLES AND TYPES OF ORGANIZATIONS

- For IT professionals, nature of employer dictates:
- How IT budgets are approved
- Speed and method of technology procurement
- Compliance and data governance frameworks
- Risk tolerance for system failures and innovations

1.1 COMMERCIAL ORGANIZATIONS

- Sole Proprietorship
- Single owner, DTI registration, unlimited liability
- Lean IT budgets, no enterprise architecture
- Consumer-grade SaaS, free social media tools for CRM
- No cybersecurity budget, no disaster recovery
- Owner's laptop crash = entire business data infrastructure down

- Partnership
- Civil Code, SEC registration, two or more individuals share profits and liabilities
- Common in law firms, accounting practices, boutique tech consultancies
- Robust Role-Based Access Control and immutable audit logs needed
- No single partner should alter financial records without oversight
- Secure collaborative knowledge management systems

- Stock Corporation
- SEC registration, RA 11232, distinct juridical personality
- Capital raised through shares, Board of Directors governance
- Enterprise-grade solutions (SAP, Oracle, Microsoft Azure)
- Automate BIR e-filing, SEC reporting, LGU permits
- Align with COBIT or ITIL for IT risk management

1.2 NON-COMMERCIAL ORGANIZATIONS

- Non-Stock Corporations, NGOs, Foundations
- SEC registration under Revised Corporation Code
- No dividends, surplus goes back to mission
- Funded by grants, donations, philanthropic endowments
- Track and display every peso of donor spending
- Transparency is currency for future funding
- Software donation programs (TechSoup Philippines), open-source, discounted NGO tiers
- Sensitive data on marginalized communities or human rights cases
- High priority on encryption and data privacy aligned with RA 10173

- Cooperatives
- CDA regulation under RA 9520
- "One member, one vote" regardless of shares held
- Prevalent in agricultural, credit, transport sectors
- Systems must calculate patronage refunds based on usage
- Localized interfaces (Tagalog, Cebuano, Ilocano)
- Optimized for low-bandwidth mobile networks

- Government Agencies and GOCCs
- 1987 Philippine Constitution, Article XI
- Funded by national budget via GAA or public revenues
- Bureaucratic, highly regulated, public and political scrutiny
- Government Procurement Reform Act (RA 9184) applies
- PhilGEPS public competitive bidding for all major IT acquisition
- Procurement can stretch months or years, technology outdated by deployment
- Must comply with DICT guidelines, GISP, interoperability frameworks
- Commission on Audit scrutiny
- Agile scope changes mid-way can be flagged as irregular expenditure

- Sector Comparison

- Commercial: highest agility, profit-driven, ROI focus, IP protection
- Non-Commercial: medium agility, mission impact, donor trust, beneficiary privacy
- Public Sector: lowest agility, compliance focus, FOI balanced with security

2.0 COMPANIES LIMITED BY SHARES AND STRUCTURE MODELS

- Stock corporations are dominant commercial technology vehicle
- Limited liability encourages high-risk innovation
- One Person Corporation (RA 11232)
  - Single developer can retain 100% ownership
  - Corporate limited liability
  - Easier to secure IP rights and contract with international clients

- Remote Governance (RA 11232)
  - Board and shareholder meetings via teleconferencing
  - Remote or absentia voting allowed
  - IT departments legally engineering corporate governance
  - Secure, auditable, authenticated portals for shareholders

- SEC Digital Enforcement
  - All filings, summons, notices via SEC-registered official email
  - IT failure to maintain channels can result in fines or contempt
  - HARBOR (Hierarchical and Applicable Relations and Beneficial Ownership Registry)
  - Systems must map and report natural persons holding 20% or more voting rights

2.1 ORGANIZATIONAL STRUCTURE MODELS

- Functional Structure (Siloed)
  - Grouped by technical capabilities (Database, Frontend, Security)
  - Prevalent in traditional conglomerates, manufacturing
  - Builds deep technical expertise
  - Severe communication silos
  - Developers write code without understanding network impact
  - Major IS risk: Deployment bottlenecks, lack of business agility

- Divisional Structure (Product/Market Focused)
  - Semi-autonomous divisions by product or geography
  - Each division has dedicated IT staff
  - Used by major telcos and fintech (separate arms for mobile money vs broadband)
  - Promotes fast product iterations, high market responsiveness
  - Duplication of resources (redundant software licensing)
  - Major IS risk: Data fragmentation, redundant software costs

- Matrix Structure (Dual Reporting)
  - Employee belongs to functional department but assigned to project
  - Reports to Functional Manager and Project Manager/Product Owner
  - Standard in IT-BPM hubs (BGC, Makati, Cebu)
  - Highly efficient for resource sharing
  - Requires strong management maturity
  - Philippine IT professionals suffer role confusion, burnout
  - Conflicting priorities between Line Manager (career) and Product Owner (sprint deadlines)
  - Major IS risk: Dual-reporting conflict, developer burnout

- Flat Structure (Decentralized Autonomy)
  - Middle management stripped away
  - Software teams have complete product autonomy
  - Embraced by early-stage FinTech, Web3, EdTech startups
  - Fosters innovation, zero bureaucratic lag
  - As startup scales, operational chaos
  - Inconsistent codebase standards, weak internal controls
  - Major IS risk: Weak internal controls, compliance audit failures

3.0 JOB DESIGN PRINCIPLES

- Job Enlargement (Horizontal Expansion)
  - Adding wider variety of tasks at same level
  - Example: manual QA tester writing automated scripts, backend developer doing frontend tweaks
  - Prevents burnout from repetitive tasks
  - Builds cross-functional redundancy

- Job Enrichment (Vertical Expansion)
  - Granting more autonomy, accountability, decision-making
  - Example: Senior Engineer designing and deploying cloud infrastructure within guardrails
  - Improves job satisfaction
  - Prepares for leadership roles (Tech Leads, Enterprise Architects)

- Job Rotation (Cross-Functional Redundancy)
  - Moving employees through different roles over time
  - Example: developer to IT Support, cybersecurity to application development
  - Vital risk-mitigation in Philippine market
  - Eliminates Critical-Man Risk (system collapses if one developer quits)
  - Builds empathy for end-users and support personnel

4.0 CENTRALIZATION, OUTSOURCING, TAKEOVERS AND MERGERS

- Centralized IT Infrastructure
  - All processing, storage, policy at HQ (typically Metro Manila)
  - Reduces licensing overhead, guarantees uniform compliance
  - One audit surface, one governance model
  - Risk: Philippine connectivity problem
  - If fiber trunkline or subsea cable breaks (typhoons), provincial branches black out

- Decentralized IT Infrastructure
  - Regional offices maintain local servers, autonomous decisions
  - Exceptional business continuity (Davao keeps trading when Manila down)
  - Local customization for local markets
  - Risk: Governance fragmentation
  - Proving national compliance across isolated silos is auditor's nightmare

- Outsourcing
  - Hiring third-party vendor for IT functions
  - Cannot outsource legal liabilities under RA 10173
  - Hiring company = Personal Information Controller (PIC) - ultimate accountability
  - Vendor = Personal Information Processor (PIP) - processes on PIC's instructions
  - SLA must hardcode:
    - AES-256 encryption at rest and in transit
    - No commingling of data on shared virtual servers
    - Vendor audit before integration
    - 24-48 hour breach notification to PIC (NPC requires 72 hours)

- Mergers, Acquisitions, Takeovers
  - Massive high-risk data and systems integration project
  - Critical roadblocks:
    - Reconciling duplicate customer/financial records without corrupting ledgers
    - Inheriting unknown vulnerabilities (unpatched systems, malware)
    - Choosing which system survives, deprecating the other
  - Philippine Competition Commission (RA 10667) monitors antitrust
  - Compulsory notification if both metrics met:
    - Ultimate parent entity Philippine revenues/assets exceeds PHP 9.1 Billion
    - Target entity assets/gross revenues exceeds PHP 3.8 Billion
  - Gun-jumping: integrating before PCC approval voids transaction with penalties
  - IS integration plan must remain partitioned until antitrust clearance

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MODULE 3: BUSINESS FINANCE AND ACCOUNTING
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1.0 FINANCING A START-UP COMPANY AND BUSINESS PLANS

- IT business plan differs from traditional businesses
- Investors look for scalability, market path, technical viability

- Anatomy of a Business Plan
  - Executive Summary: problem, solution, funding ask (one page)
  - Market Analysis: TAM, SAM, SOM
  - Revenue Model: how software makes money
  - Technical Feasibility and Roadmap: MVP to scaled architecture

- Revenue Models
  - SaaS: recurring subscription, predictable but needs high retention
  - Freemium: free core, paid premium, needs massive user scale
  - Licensing/White-labeling: selling underlying technology to other businesses

- Capital Requirements
  - Software start-ups typically operate at loss early
  - Focus on calculating financial runway
  - CapEx vs OpEx distinction is fundamental shift
  - Historically: massive upfront capital for physical infrastructure

  - Today: cloud computing turned infrastructure into utility bill
  - Underestimating development time is number one reason start-ups run out of cash
  - Include buffer for technical debt and bug fixing

- Funding Sources
  - Bootstrapping: full equity retention, complete control, high personal risk, slower growth
  - Crowdfunding: validates demand, no equity lost, all-or-nothing risk, massive marketing
  - Angel Investors: mentorship, patient capital, loss of equity, difficult match
  - Venture Capital: massive capital, growth support, high pressure, loss of control
  - Bank Loans: keep equity, fixed repayment, difficult without collateral

- No single correct path
  - Early-stage: Bootstrapping or Crowdfunding for MVP
  - After validation: Angel Investors or VC for scale, or Bank Debt if revenue is predictable

- Assessing Financial and Project Risk
  - Business plan is tool for building trust
  - Investors buy into sustainable financial engine, not just code
  - Show problem is expensive enough for multi-year contract
  - Know cash needed for 6-12 month B2B sales cycle
  - Understand CAC and compliance hurdles

2.0 FINANCIAL ACCOUNTING FOR IT PROFESSIONALS

- Financial Accounting: standardized reporting to external stakeholders (investors, government)

- Profit and Loss Statement (Income Statement)
  - "Video" showing performance over a period (quarter/year)
  - Revenue: SaaS fees, licenses, consulting hours
  - COGS: direct costs to deliver software (cloud hosting, API fees, customer support)
  - OpEx: running business costs (developer salaries, rent, Slack/Jira licenses)
  - Net Profit: what remains after all expenses and taxes

- Gross Revenue
  - Absolute "top line" - total unadjusted money from goods/services

- Gross Profit
  - Revenue minus COGS
  - First measure of profitability

- Gross Margin
  - (Revenue minus COGS) divided by Revenue times 100
  - Investors pay for margin, not revenue
  - Example: Hardware company 30% margin vs SaaS company 85% margin
  - SaaS keeps more profit to invest in growth

- Balance Sheet
  - "Photograph" on a single day (usually last day of year)
  - Shows what company owns and owes
  - Assets = Liabilities + Equity
  - Assets: cash, laptops, servers, capitalized software
  - Liabilities: bank loans, unpaid cloud invoices, deferred revenue
  - Equity: VC money, founder shares, retained earnings

- Capitalization
  - Routine bug fixes = expense on P&L
  - Building new patentable software platform = capitalizable as asset on Balance Sheet

- Deferred Revenue
  - Client pays 120,000 upfront for 12-month license
  - Cannot claim all 120,000 as revenue today
  - Record as liability (owe 12 months of service)
  - Recognize 10,000 revenue each month

- Cash Flow Forecasting
  - Estimating timing and amount of money moving in and out
  - Strictly liquidity management tool
  - Primary purpose: predict runway
  - Determine when company runs out of money
  - Secure funding or adjust spending before crisis

- Profit vs Cash
  - Profit: surplus after expenses, accrual accounting, revenue recorded when contract signed, not when paid
    - Indicates if business model is viable
  - Cash: actual money in bank on any given day
    - Indicates if business can survive current week/month
  - Dilemma: 90-day payment terms from enterprise clients

- Profitable on paper but zero cash for three months
- Cash flow forecasting manages the 90-day gap

3.0 MANAGEMENT ACCOUNTING

- Financial Accounting: looking backward, reporting score to external investors
- Management Accounting: looking forward, helping internal leaders run playbook

- Cost Center vs Profit Center
  - Cost Center: no direct revenue, costs money to run (HR, Legal, traditional IT)
    - Goal: minimize expenses without breaking system
  - Profit Center: brings in more money than spends
    - Two ways IT becomes profit center:
      - Selling proprietary data/software to outside clients (external)
      - Billing other internal departments (internal chargeback)

- Direct Costs
  - One-to-one relationship with specific project
  - If project cancelled, costs vanish
  - Direct Labor: full-time engineer's salary allocated proportionally
  - Contracted Labor: external contractors hired exclusively for project
  - Dedicated Infrastructure: cloud testing environments, premium SDK/API licenses

- Indirect Costs
  - Shared operational foundation of company
  - If project cancelled, costs do not change
  - Shared Infrastructure: GitHub Enterprise, Slack, Jira, Notion
  - Shared Human Capital: DevOps team, IT Helpdesk, CIO, legal team
  - Facilities and Administrative: office rent, security, hardware depreciation, recruiting

- Overhead Absorption
  - Indirect costs must be absorbed into individual project costs
  - If project doesn't cover share of overhead, company loses money
  - Choose allocation base (usually developer hours)
  - Example: P2,000,000 overhead / 10,000 hours = P200/hour
  - App A (6,000 hours) absorbs P1,200,000
  - App B (4,000 hours) absorbs P800,000
  - Project manager must include direct salaries AND overhead in estimates
  - Pricing only on direct salaries means company can't pay rent

- IT Chargeback Model
  - IT bills other departments for technology they consume
  - Prevents departments from treating IT resources as "free"
  - Marketing requests database only if truly needed when they see the cost
  - Track usage: laptops issued, server storage used, development hours
  - Monthly internal invoice based on direct costs plus absorbed overhead

4.0 INVESTMENT APPRAISAL

- Investment Appraisal (Capital Budgeting)
  - Structured evaluation of proposed long-term project
  - Answers: will future financial returns justify upfront cost and risk

- Time Value of Money (TVM)
  - Money today is worth more than same amount in future
  - Three core factors:
    - Opportunity Cost: money today can be invested for returns
    - Inflation: purchasing power drops over time
    - Risk and Uncertainty: future cash may not materialize
  - Compounding: moving forward to Future Value
    - FV = PV x (1 + r)^n
  - Discounting: pulling backward to Present Value
    - PV = FV / (1 + r)^n

- Without TVM, dangerous mistakes happen
  - Example: Cost 4M, saves 1M/year for 4 years
  - Naive conclusion: breaks even (4M in, 4M out)
  - Reality: Year 3 and 4 savings worth less due to TVM
  - When discounted, project actually loses money

- Payback Period
  - First financial filter for IT projects
  - Answers: how long until we get our money back
  - Favored by executives for addressing operational question

- Tech sector prefers shorter payback due to rapid change
- Constant Annual Cash Flows: Initial Investment / Annual Cash Inflow
- Variable Cash Flows: Cumulative Cash Flow Method

- Discounted Cash Flow (DCF) and Net Present Value (NPV)
- Gold standard for investment appraisal
- Answers: accounting for TVM, cost of capital, and risk, will this project make company richer or poorer
- Does not treat future cash flows as absolute truths
- Two hard realities:
  - Opportunity Cost of Capital: money used here can't be used elsewhere
  - Delayed Gratification is Worth Less: peso earned in Year 5 worth less than today

- NPV Calculation
- Discount each future cash flow back to Present Value
- Sum of discounted inflows minus initial investment
- NPV = sum of CFt / (1+r)^t minus Initial Investment
- CFt = net cash inflow in specific year
- r = discount rate (hurdle rate, cost of capital)
- t = specific year
- n = project lifespan

- Three Decision Rules
- NPV > 0: recovers cost, meets required return, creates wealth
  - Decision: ACCEPT
- NPV = 0: meets required return, creates nothing extra
  - Decision: INDIFFERENT (accept only for strategic/compliance reasons)
- NPV < 0: fails required return, destroys company wealth
  - Decision: REJECT

- Tech Project Risk Profiles
- WACC as default discount rate for corporate finance
- IT projects carry high execution risks
- Companies apply Risk Premium to discount rate
- Standard: 8%, High-risk software: 12-15%
- Higher discount rate penalizes future cash flows more aggressively
- Forces strong early returns for positive NPV
- Cap tech project horizons to 3-5 years maximum
- Rapid technological change makes longer horizons risky

- Key Takeaway
- Tech projects rarely fail because code didn't compile
- They fail because they run out of cash, blow budgets, or can't prove economic value
- Brilliant code is worthless if business goes bankrupt waiting for payment
- Mastering financial architecture makes you strategic engine, not just expense

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SYNTHESIS OF MODULES 1-3
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- Three modules are not separate topics
- They are three lenses that must be used at the same time

- Law Lens
- What must we do to stay out of court?
- What should we do to be ethical?
- Governance, compliance, codes of conduct
- Criminal vs civil liability
- Data Privacy Act, regulatory frameworks

- Structure Lens
- Who decides, who reports to whom?
- Where does the control break?
- Organizational models, job design
- Centralization, outsourcing, mergers
- Legal and operational nature of organizations

- Money Lens
- Does the project create wealth?
- After discounting time and risk?
- Business plans, funding sources
- Financial accounting, management accounting
- Chargeback models, TVM, payback, NPV

- AlagaVerse Tech Inc. Case Study
- FinTech + HealthTech hybrid startup, BGC, Taguig
- Micro-insurance, emergency funding, AI credit scoring
- Serves unbanked: delivery riders, BPO workers, micro-entrepreneurs

  - Grew from 12 informal developers to 50+ engineers
  - Series A closed, IPO target late 2026

- PadalaCare Project
  - Automated real-time micro-loan and health-funding module
  - Ingests alternative user data (mobile load telco data, delivery rider transactions)
  - Instantly disburses emergency cash using predictive modeling

- The Board Decision
  - Option A: Outsource to Cebu-based vendor
    - Upfront: 4,000,000
    - Annual savings: 1,800,000 for 3 years
    - Ships in: 4 months
  - Option B: Build in-house
    - Upfront: 6,000,000
    - Annual savings: 2,500,000 for 3 years
    - Ships in: 9 months
  - Hurdle rate: 12% (high-risk software)
  - Horizon: 3 years maximum

- Financial Analysis
  - Discount factors at 12%
  - PV of Year 1 savings = Annual savings x 0.893
  - PV of Year 2 savings = Annual savings x 0.797
  - PV of Year 3 savings = Annual savings x 0.712
  - Total PV = sum of discounted inflows
  - NPV = Total PV - Initial Investment
  - Payback = Initial Investment / Annual Cash Inflow

- Legal Analysis
  - Under RA 10173, outsourcing does not transfer liability
  - AlagaVerse remains Personal Information Controller (PIC)
  - Vendor is Personal Information Processor (PIP)
  - SLA must hardcode: encryption, no commingling, 24-48 hour breach notification
  - 24-48 hours gives AlagaVerse time to meet NPC's 72-hour reporting window

- Structural Analysis
  - Can flat startup structure adequately control vendor?
  - Segregation of duties, internal controls
  - Critical-man risk
  - Ability to audit vendor

- Three Vetoes
  - Legal Veto: who holds liability? Is it merely compliant or right?
  - Structural Veto: can org chart run this without control breaking?
  - Financial Veto: does it create wealth after discounting time and risk?

- Key Takeaway
  - Decision that passes only one test is not professional decision
  - If answer changes depending on which lens you look through first, you have found the real decision
  - All three lenses must be satisfied
  - The law tells you the minimum you can get away with
  - Your professional code tells you what you owe the public
  - Neither can be delegated to anyone else

QUIZ SAMPLE 1

PCIS2301 - Professional Issues in Information Systems
Module 1-3 Comprehensive Multiple Choice Questions

Questions 1-10: Law and the Computing Profession (Module 1)

1. You are the newly appointed CTO of a Philippine FinTech startup. During a board meeting, the CEO asks you to explain the difference between statutory law and common law in the context of IT regulation. Which of the following responses correctly distinguishes these two sources of law?

A) Statutory law is created by judges through court decisions, while common law is written by legislative bodies

B) Statutory law is formally enacted by legislatures (like RA 10173), while common law develops through judicial precedent when judges rule on novel tech cases

C) Statutory law only applies to criminal matters, while common law only applies to civil disputes

D) Statutory law is enforced by regulatory agencies, while common law is enforced by professional bodies like BCS

---

2. AlagaVerse Tech Inc. is developing an AI-driven credit scoring system that processes alternative user data such as mobile load telco data and delivery rider transaction frequencies. During the Committee and Report Stage of a proposed law regulating AI in financial services, the government requests technical expertise. Which stage of the legislative pipeline does this represent, and why are computing professionals crucial at this point?

A) First Reading - because the bill is formally announced to the public

B) Second Reading - because lawmakers debate the overarching principles of the bill

C) Committee and Report Stage - because line-by-line review requires technical experts to prevent unfeasible laws

D) Proposal Stage - because the White Paper needs technical validation before legislation is drafted

---

3. AlagaVerse outsources the development of its PadalaCare micro-loan feature to a Cebu-based vendor. Under the Data Privacy Act of 2012 (RA 10173), which of the following correctly describes the legal relationship between AlagaVerse and the vendor?

A) Both AlagaVerse and the vendor are equally liable as co-controllers of personal data

B) AlagaVerse is the Personal Information Processor (PIP), and the vendor is the Personal Information Controller (PIC)

C) AlagaVerse is the Personal Information Controller (PIC), and the vendor is the Personal Information Processor (PIP), with AlagaVerse retaining ultimate legal accountability

D) The vendor assumes all legal liability once the outsourcing contract is signed, freeing AlagaVerse from data protection obligations

---

4. A software engineer at AlagaVerse discovers that the automated recruitment AI system they are building is inherently biased against older applicants. The engineering manager instructs them to ignore the issue to meet the project deadline. Under the BCS Code of Conduct, which pillar requires the engineer to raise this concern?

A) Duty to the Relevant Authority (Employers and Clients)

B) Professional Competence and Integrity

C) The Public Interest - which prioritizes public safety, privacy, and human rights over employer wishes

D) Duty to the Profession

---

5. The National Privacy Commission (NPC) audits AlagaVerse and finds that while the company has a Privacy Manual, it has not conducted a Privacy Impact Assessment or appointed a Data Protection Officer. How many of the NPC's 5 Pillars of Compliance has AlagaVerse failed to meet?

A) 1 pillar - only missing the Data Protection Officer

B) 2 pillars - missing both Privacy Impact Assessment and Data Protection Officer

C) 3 pillars - missing Privacy Impact Assessment, Data Protection Officer, and Privacy Manual

D) The company has met all pillars since it has a Privacy Manual

---

6. A hacker breaches AlagaVerse's systems and steals customer data. Under the Cybercrime Prevention Act of 2012 (RA 10175), which of the following activities is specifically criminalized?

A) Failure to appoint a Data Protection Officer

B) Unauthorized access to computer systems, data interference, and identity theft

C) Non-compliance with ISO 27001 certification requirements

D) Failure to report a data breach within 72 hours

---

7. The IT Director of AlagaVerse is considering implementing an internal ethics board to review AI bias before launching new features. The legal team reminds them that they must also ensure the company's website displays proper cookie consent banners to comply with data protection regulations. This scenario illustrates the distinction between:

A) Criminal law and civil law

B) Governance (proactive, ethics-driven) and Compliance (reactive, legally mandated)

C) Statutory law and common law

D) Centralized and decentralized IT infrastructure

---

8. AlagaVerse launches a mobile app with a known security vulnerability to meet a marketing deadline. A data breach occurs, and the company faces both criminal prosecution by the state and a civil lawsuit from affected customers. Which of the following correctly describes the standards of proof required in these two legal actions?

A) Criminal: Balance of probabilities; Civil: Beyond a reasonable doubt

B) Criminal: Beyond a reasonable doubt; Civil: Balance of probabilities

C) Both require proof beyond a reasonable doubt

D) Both require proof on the balance of probabilities

---

9. You are a senior developer at AlagaVerse and discover that a colleague is copying entire blocks of proprietary, copyrighted code from a competitor's system. Under the BCS Code of Conduct, which pillar requires you to address this code plagiarism?

A) The Public Interest

B) Professional Competence and Integrity

C) Duty to the Relevant Authority

D) Duty to the Profession - because it damages the integrity and reputation of the entire IT industry

---

10. A Filipino IT professional has been working as a developer for 10 years but has not engaged in any Continuing Professional Development (CPD) activities. They design a cloud-based medical records system using outdated security methodologies, leading to a major data breach. Under the BCS Code of Conduct, which pillar makes this a disciplinable offense?

A) The Public Interest - they failed to protect public health

B) Professional Competence and Integrity - CPD is an enforceable duty to maintain competence

C) Duty to the Relevant Authority - they failed their employer

D) Duty to the Profession - they damaged the profession's reputation

---

Questions 11-20: Organizational Structure and Management (Module 2)

11. AlagaVerse Tech Inc. started as a garage startup with 12 informal developers and has now grown to 50+ engineers. The CEO demands a complete corporate restructuring. The company operates in the Philippine FinTech and HealthTech sectors. Based on the organizational models discussed, which structure is AlagaVerse currently using, and what is its primary risk?

A) Functional Structure - risk of severe deployment bottlenecks

B) Matrix Structure - risk of dual-reporting conflict and burnout

C) Flat Structure - risk of weak internal controls and failed external audits

D) Divisional Structure - risk of data fragmentation

---

12. AlagaVerse is preparing for an IPO in late 2026. The Board of Directors is concerned about the company's ability to pass external IT audits. As CTO, you recommend a restructuring that properly separates duties between development, quality assurance, and cybersecurity. Which principle of organizational design does this primarily address?

A) Job enlargement

B) Job enrichment

C) Segregation of duties and internal controls

D) Job rotation

---

13. AlagaVerse has a critical-man risk problem - only one developer understands the core payment processing system. The CEO is concerned about what would happen if this developer leaves. Which job design strategy would best mitigate this risk?

A) Job Enlargement - adding more tasks to the developer's role

B) Job Enrichment - giving the developer more decision-making authority

C) Job Rotation - systematically moving developers through different roles to build cross-functional redundancy

D) Outsourcing - hiring a vendor to manage the payment system

---

14. The Philippine Competition Commission (PCC) is reviewing AlagaVerse's acquisition of a smaller logistics tech startup valued at PHP 4 Billion. The IT team begins integrating the two companies' databases and migrating customer data before receiving PCC approval. What is this practice called, and what is the consequence?

A) Due diligence - legally required before merger completion

B) Gun-jumping - the transaction can be legally voided with financial penalties

C) System integration - standard practice in any merger

D) Data migration - acceptable if both parties agree

---

15. AlagaVerse is considering centralizing all IT infrastructure in its BGC headquarters versus maintaining decentralized systems in regional offices. A major typhoon hits Metro Manila, and the central data center goes offline. The decentralized branches in Cebu and Davao continue operations. This scenario illustrates which trade-off in IT infrastructure design?

A) Centralization guarantees better security compliance but sacrifices operational availability

B) Decentralization provides better business continuity but creates governance fragmentation challenges

C) Both models provide equal resilience to natural disasters

D) Centralization is always superior for Philippine businesses due to regulatory requirements

---

16. The Board of AlagaVerse is debating whether to outsource software development to a vendor. The legal counsel reminds the board that under RA 10173 (Data Privacy Act), if they outsource data processing, what is the legal status of AlagaVerse and the vendor?

A) AlagaVerse becomes a Personal Information Processor (PIP), eliminating their liability

B) AlagaVerse remains the Personal Information Controller (PIC) and retains ultimate legal accountability

C) The vendor becomes the sole liable party under the outsourcing contract

D) Both companies share equal liability as joint controllers

---

17. AlagaVerse's IT department is structured functionally, with separate teams for Frontend, Backend, Database, and Security. A new feature requires coordination across all teams, but communication is slow and deployments are frequently delayed. This is a classic example of which organizational risk?

A) Matrix structure role conflict

B) Flat structure weak controls

C) Functional structure deployment bottlenecks and silo communication problems

D) Divisional structure data fragmentation

---

18. A Philippine government agency is developing a public health information system. Unlike AlagaVerse (a private FinTech), this agency must follow strict procurement rules under RA 9184. What is the primary implication for the IT project?

A) The agency can procure technology faster than private companies due to government resources

B) IT procurement must go through public competitive bidding via PhilGEPS, often stretching timelines to months or years

C) The agency is exempt from data privacy regulations

D) The agency can use any vendor without competitive bidding

---

19. AlagaVerse is moving from a flat organizational structure to a more formal structure. The IT Director proposes a matrix structure where engineers report to both a functional manager and a project manager. In the Philippine IT-BPM context, what is the primary risk of this structure?

A) High software licensing costs

B) Data fragmentation across divisions

C) Dual-reporting conflict and developer burnout

D) Inability to innovate quickly

---

20. A non-stock, non-profit organization developing an open-source health tracking app for marginalized communities has very limited funding. Which of the following best describes the IT resourcing reality for this type of organization in the Philippines?

A) They can afford enterprise-grade solutions due to government subsidies

B) They rely heavily on software donation programs, open-source software, and discounted NGO tiers

C) They have unlimited access to commercial cloud services

D) They are exempt from data privacy compliance

Questions 21-30: Business Finance and Accounting (Module 3)

21. AlagaVerse must decide between outsourcing the PadalaCare credit-scoring engine (Option A: P4,000,000 upfront, P1,800,000 annual savings for 3 years) or building it in-house (Option B: P6,000,000 upfront, P2,500,000 annual savings for 3 years). Using a 12% discount rate, which statement is true?

A) Both options have positive NPV and should be accepted

B) Option A has a higher NPV than Option B

C) Option B has a higher NPV than Option A

D) Both options have negative NPV and should be rejected

---

22. Using the same AlagaVerse scenario above, what is the payback period for Option A (outsourcing) if savings are constant each year?

A) 2.00 years

B) 2.22 years (P4,000,000 ÷ P1,800,000)

C) 2.40 years (P4,000,000 ÷ P1,666,667 average discounted savings)

D) 3.00 years

---

23. AlagaVerse's finance team analyzes the PadalaCare project. The upfront cost is P4,000,000, and it generates P1,800,000 in annual savings. Using a 12% discount rate with a 3-year horizon, calculate the Net Present Value (NPV) for Option A. Year 1 factor=0.893, Year 2=0.797, Year 3=0.712.

A) P10,800,000 (positive)

B) P5,400,000 (positive)

C) Approximately P323,600 (positive) - (P1.607M + P1.435M + P1.282M) - P4.000M

D) Negative, because annual savings don't exceed the initial investment

---

24. The CFO of AlagaVerse explains to the board that although the PadalaCare project shows a positive profit on paper (P1,800,000 annual savings × 3 years = P5,400,000, exceeding the P4,000,000 investment), an enterprise client using the service has 90-day payment terms. Which concept does this illustrate?

A) Gross margin is more important than net profit

B) Profit is an accounting measure (accrual basis), while cash is the actual money in the bank (liquidity)

C) The project should be rejected regardless of accounting profit

D) Deferred revenue should be recorded as an asset

---

25. AlagaVerse is preparing its financial statements. The company received P120,000 upfront for a 12-month software license. Under proper accounting treatment, which statement is correct?

A) The entire P120,000 should be recorded as revenue immediately

B) P120,000 should be recorded as deferred revenue (a liability) and recognized as P10,000 revenue each month

C) The amount should be recorded as an asset and depreciated over 12 months

D) The amount should be recorded as equity

---

26. The IT department at AlagaVerse incurs P2,000,000 in indirect overhead (rent, management salaries, shared tools). Developers log 10,000 total hours. App A takes 6,000 hours, and App B takes 4,000 hours. Under absorption costing, how much overhead should be charged to App A?

A) P1,200,000 (6,000 hours × P200/hour)

B) P800,000 (4,000 hours × P200/hour)

C) P2,000,000 (all overhead charged to App A)

D) P0 (overhead cannot be allocated to specific projects)

---

27. AlagaVerse is considering implementing an IT chargeback model. What is the primary purpose of this system?

A) To generate external revenue for the IT department

B) To make other departments treat IT resources as valuable by billing them for consumption, preventing waste

C) To eliminate the IT department's budget entirely

D) To comply with SEC reporting requirements

---

28. Two companies generated P60,000,000 in revenue: Company A (hardware) has COGS of P42,000,000, and Company B (SaaS) has COGS of P9,000,000. Which statement about their gross margins is correct?

A) Company A has a higher gross margin (70%) than Company B (15%)

B) Company B has a higher gross margin (85%) than Company A (30%)

C) Both companies have the same gross margin

D) Gross margin cannot be calculated with the given information

---

29. AlagaVerse's CFO is evaluating a software project with a 3-year life. The discount rate is 12%, representing the company's cost of capital plus a risk premium for high-risk software development. Which concept does this illustrate about tech project appraisal?

A) A higher discount rate makes future cash flows more valuable

B) A higher discount rate penalizes future cash flows more aggressively, requiring strong early returns for positive NPV

C) The discount rate doesn't affect NPV calculations for short-term projects

D) The discount rate only applies to projects with borrowed capital

---

30. An enterprise SaaS startup signed a multi-million peso contract today, showing a massive profit on paper. However, the client has 90-day payment terms, and the startup has only 60 days of cash runway. Which financial tool is most critical for managing this gap?

A) Balance sheet

B) Cash flow forecasting - to predict when the company will run out of money

C) Profit and Loss statement

D) Gross margin analysis

Questions 31-40: Synthesis and Integration

31. AlagaVerse Tech Inc. is at a board meeting deciding whether to outsource or build in-house the PadalaCare credit-scoring engine. As CTO, you must consider three lenses: Law, Structure, and Money. A decision that passes only one test is considered:

A) Acceptable if it passes the financial test

B) Acceptable if it passes the legal test

C) Not a professional decision - all three vetoes must be satisfied

D) Acceptable if the board majority votes for it

---

32. The board of AlagaVerse is divided on the PadalaCare project. The DPO (Data Protection Officer) argues that outsourcing does not transfer liability, the CFO presents NPV calculations, and the CTO questions whether a flat startup structure can adequately control the vendor. This scenario illustrates the integration of which three modules?

A) Marketing, Operations, and Human Resources

B) Law (Module 1), Structure (Module 2), and Money (Module 3)

C) Finance, Accounting, and Ethics

D) Software Engineering, Database Management, and Cybersecurity

---

33. AlagaVerse's Board must vote on the PadalaCare project. The CFO calculates that Option A (outsourcing) has a positive NPV of P323,600, while Option B (in-house) has a negative NPV. However, the DPO warns that outsourcing data processing to a vendor carries significant liability under RA 10173. The CTO notes that the flat organizational structure may not have adequate controls for vendor management. What should the board's decision be?

A) Accept Option A because NPV is positive

B) Accept Option B because in-house development gives better control

C) Reject both options until legal and structural concerns are addressed

D) Accept Option A with conditions to ensure legal compliance and structural controls

---

34. AlagaVerse's IT Director is presenting a CPD (Continuing Professional Development) plan to the Board. They propose attending structured training courses, reading academic journals, mentoring junior developers, and participating in industry conferences. Which model of CPD does this balanced approach most closely represent?

A) Deficit model - targeting individual weaknesses

B) Training model - acquiring predefined technical skills

C) Emancipatory model - integrating multiple approaches with professional ownership over learning

D) Standard-based model - demonstrating competence against rigid standards

---

35. AlagaVerse is considering acquiring a smaller FinTech startup. The acquisition triggers mandatory notification to the Philippine Competition Commission (PCC) because the ultimate parent entity has PHP 10 Billion in Philippine assets and the target entity has PHP 4 Billion in assets. Under RA 10667, what happens if the IT team begins system integration before PCC approval?

A) The integration is allowed as long as both companies agree

B) The transaction is legally voided with potential financial penalties (gun-jumping)

C) PCC approval is only required for mergers of public companies

D) The companies can proceed if they notify PCC within 30 days of integration

---

36. A software engineer at AlagaVerse discovers that their manager is pressuring them to launch a feature despite known security vulnerabilities. The engineer is a member of BCS and bound by its Code of Conduct. The manager's request violates which pillar, and what is the engineer's professional obligation?

A) Duty to the Relevant Authority - the engineer must comply with the manager

B) Professional Competence and Integrity - the engineer must only accept work they are qualified for

C) The Public Interest - the engineer must prioritize public safety and must refuse or warn in writing

D) Duty to the Profession - the engineer must report the manager to BCS immediately

---

37. A Philippine IT professional has a "job" as a developer but is not a member of any professional body. A colleague who is a Chartered IT Professional (CITP) through BCS has additional responsibilities and is held to higher standards. According to the module, what fundamentally differentiates a "job" from a "profession"?

A) Salary level determines professional status

B) A profession requires specialized knowledge, rigorous entry standards, self-governance, and duty to the public over profit

C) Only government employees can be considered professionals

D) A profession requires a master's degree

---

38. AlagaVerse's legal team is drafting an outsourcing contract for the PadalaCare project. Under RA 10173, the Service Level Agreement (SLA) must explicitly require the vendor to maintain specific security measures. Which of the following is NOT a requirement that should be in the SLA?

A) AES-256 encryption for data at rest and in transit

B) No commingling of AlagaVerse's data with other clients' data

C) Breach notification to AlagaVerse within 24-48 hours

D) The vendor assumes all legal liability for any data breach

---

39. A government IT manager is developing a critical system and must follow the Government Procurement Reform Act (RA 9184). They cannot simply choose the best vendor; they must go through public bidding via PhilGEPS. Which concept from the organizational typology section does this illustrate?

A) The Public Sector has the highest system agility

B) The Public Sector faces the tightest constraints and lowest system agility due to rigid procurement laws

C) Government agencies are exempt from data privacy compliance

D) Public sector IT projects face no budget limitations

---

40. AlagaVerse is calculating the cost of building a new software feature. The developer's salary is allocated based on time spent on the project (60% on Project A, 40% on Project B), and shared costs like GitHub Enterprise and office rent must be absorbed into project costs. This scenario illustrates the distinction between:

A) OpEx and CapEx

B) Direct costs (can be allocated proportionally) and Indirect costs (must be absorbed using a rate)

C) Profit and Cash

D) Assets and Liabilities

Questions 41-50: Advanced Application

41. AlagaVerse's CFO presents the following NPV calculations for the PadalaCare project at a 12% discount rate: Option A shows a positive NPV of approximately P323,600, while Option B (in-house) shows a negative NPV. Given a 3-year project horizon, what should the board conclude?

A) Option A creates company wealth and should be accepted

B) Both options destroy company wealth

C) The project horizon should be extended to 5 years to make Option B viable

D) The board should focus only on payback period, not NPV

---

42. Under the Philippine government's typology, a GOCC (Government-Owned and Controlled Corporation) like GSIS must balance transparency (Freedom of Information) with national security and citizen privacy. This creates what unique challenge for IT professionals in the public sector?

A) No challenge - private sector standards apply equally

B) Systems must manage the delicate balance between open governance requirements and privacy/security obligations

C) GOCCs are exempt from data privacy compliance

D) GOCCs face no audit scrutiny

---

43. A developer at AlagaVerse has been systematically moved through different technical roles - backend development, database administration, and QA - over 18 months. This practice is primarily designed to:

A) Reduce salary costs

B) Eliminate critical-man risk where the system collapses if one developer quits

C) Give developers more decision-making authority

D) Comply with SEC regulations

---

44. AlagaVerse's IT department is transitioning from a cost center to a profit center. Currently, it is a "black hole" that consumes budget without generating measurable value. Which of the following strategies would NOT contribute to this transition?

A) Selling proprietary software to external clients

B) Implementing an internal chargeback model where IT bills other departments

C) Billing internal departments based on their technology consumption

D) Increasing the IT department's operating budget without accountability

---

45. The National Privacy Commission (NPC) requires breach notifications within 72 hours of discovery. When AlagaVerse outsources data processing, the Service Level Agreement (SLA) with the vendor must obligate the vendor to notify AlagaVerse within 24-48 hours. Why is the shorter notification window in the SLA necessary?

A) To give AlagaVerse time to ignore the breach

B) To ensure AlagaVerse has sufficient time to investigate and report to the NPC before the 72-hour deadline

C) It's optional and not legally required

D) The vendor should report directly to the NPC, not to AlagaVerse

---

46. A Filipino software engineer with 15 years of experience has never engaged in CPD activities. They rely entirely on skills learned in university. The CTO is considering whether this constitutes a violation of professional standards. Under Bott & Taylor's CPD framework, which statement is most accurate?

A) CPD is optional for experienced professionals

B) The computing profession is uniquely volatile - a degree from 15 years ago is structurally obsolete without CPD

C) Industry experience replaces the need for formal CPD

D) CPD only matters for junior developers

---

47. AlagaVerse's Board is reviewing the organizational structure proposed by the CTO. The proposal separates development, QA, and security into distinct reporting lines, creates a Risk and Data Governance Committee, and establishes clear escalation paths. This restructuring primarily addresses:

A) Financial reporting requirements

B) Internal controls and segregation of duties to prepare for external audits and IPO

C) Reducing the IT budget

D) Complying with international software licensing

**48. A Philippine cooperative (regulated by CDA under RA 9520) is implementing a new ERP system. Unlike a stock corporation that calculates dividends per share, this cooperative

must calculate patronage refunds based on how much members utilized the cooperative's services. This illustrates which concept?**

A) Commercial organizations have simpler IT requirements

B) The constitutional nature of the organization dictates specific system requirements and configurations

C) All organizations follow the same accounting standards

D) Cooperatives are exempt from data privacy compliance

---

49. AlagaVerse is evaluating PadalaCare using the TELOS framework. Which of the following is NOT a dimension of the TELOS feasibility framework?

A) Technological feasibility

B) Economic feasibility

C) Legal feasibility

D) Marketing feasibility

---

50. The BCS Code of Conduct has four pillars, while the ACM Code has 24 imperatives organized into four core sections. Both emphasize that professionals must move beyond asking "Is this system legal?" to asking "Is this system right?" Which concept does this represent?

A) Compliance is the ceiling, not the floor

B) Compliance is the floor, not the ceiling - ethics and governance require going beyond minimum legal requirements

C) Legal compliance is more important than ethics

D) Ethics and law are identical

Questions 51-60: Final Integration and Critical Thinking

51. AlagaVerse is considering outsourcing PadalaCare development to a Cebu-based vendor. The CTO must draft an SLA that addresses security, breach notification, and data handling. The DPO reminds the board that under RA 10173, AlagaVerse cannot outsource its legal liability. Which statement best captures the legal relationship?

A) The vendor (PIP) assumes all liability for data breaches

B) AlagaVerse (PIC) retains ultimate legal accountability, while the vendor (PIP) must process data strictly on the PIC's instructions

C) Both companies share equal liability as joint controllers

D) The NPC regulates only the PIC, not the PIP

---

52. The Board of AlagaVerse is meeting to decide on PadalaCare. The CFO presents Option A (outsource) with positive NPV, the DPO warns that liability remains with AlagaVerse, and the CTO notes that the flat organizational structure may struggle to control the vendor. What should the Board conclude about the relationship between the three lenses?

A) A positive NPV overrides legal and structural concerns

B) Legal compliance is the only thing that matters

C) If the answer changes depending on which lens you look through first, you have found the real decision - all three must be satisfied

D) Organizational structure is irrelevant to project decisions

---

**53. AlagaVerse's IT Director is designing a CPD plan. They include: attending AWS certification training (formal learning), reading security blogs and research papers (informal learning), mentoring junior developers (work-based learning), and speaking at a tech

conference (professional activity). Which statement about CPD is most accurate?**

A) Only formal learning counts as CPD

B) CPD is a balanced portfolio of all four activity types, and under BCS Pillar 2 it is an enforceable duty

C) CPD is optional for IT professionals in the Philippines

D) CPD only applies to senior management

---

54. A bank in Manila experiences a complete central data center outage due to a fire. However, its decentralized regional branches in Cebu continue processing local transactions offline. This illustrates which infrastructure trade-off?

A) Centralization guarantees better security but sacrifices operational availability

B) Decentralization provides exceptional business continuity but creates governance fragmentation challenges

C) Both models provide the same level of disaster recovery

D) The bank should completely centralize all operations

---

55. AlagaVerse's CEO wants to understand why investors value CloudSync (an enterprise SaaS company) more than Tech-Gear Inc. (a hardware company) even though both generated P60,000,000 in revenue. CloudSync's COGS is P9,000,000 (85% gross margin) while Tech-Gear's COGS is P42,000,000 (30% gross margin). Which statement best explains why investors pay for margin, not revenue?

A) Hardware companies have lower operating costs than software companies

B) Software companies have higher gross margins and can invest more in growth, making them more scalable

C) Revenue is irrelevant to investors

D) Both companies are equally attractive investments

---

56. AlagaVerse is developing a feature that ingests alternative user data (mobile load telco data, delivery rider transactions) for credit scoring. Which Philippine law is most directly applicable to this data processing activity?

A) Electronic Commerce Act (RA 8792) - governing electronic transactions

B) Data Privacy Act (RA 10173) - protecting personal data and requiring consent and security measures

C) Intellectual Property Code (RA 8293) - protecting software copyright

D) DICT Act (RA 10844) - establishing ICT policy

---

57. A developer at AlagaVerse notices that the automated recruitment AI system is biased against older applicants. When they raise the concern, the project manager tells them to ignore it to meet the deadline. If the developer is a BCS member, which of the following best describes their professional obligation?

A) Comply with the project manager to preserve employment

B) Under the Public Interest pillar, they are ethically obligated to raise the alarm and refuse to continue

C) Report the incident only after the project is complete

D) There is no obligation because AI bias is not covered by the code

---

**58. AlagaVerse is transitioning from a flat structure to a more formal structure as it prepares for IPO. The CTO recommends separating QA from Development, segregating duties between infrastructure and security

teams, and creating clear reporting lines. What is the primary driver for this restructuring?**

A) Reducing the number of employees

B) Preparing for institutional scale and external audit requirements

C) Decreasing software development speed

D) Complying with international standards that don't apply in the Philippines

---

59. Two companies are merging: a commercial bank and a FinTech startup. The IT integration team must reconcile databases with overlapping customer records, decide which core system survives, and ensure they don't integrate before PCC approval. Which Philippine law governs the antitrust review of this merger?

A) Revised Corporation Code (RA 11232)

B) Philippine Competition Act (RA 10667)

C) Data Privacy Act (RA 10173)

D) DICT Act (RA 10844)

---

60. AlagaVerse is preparing its business plan for Series B funding. The CTO needs to clearly articulate the problem, the tech-based solution, and the funding ask. According to the module, which component of a tech business plan is most critical for demonstrating understanding of the market opportunity?

A) Executive summary

B) Market analysis showing TAM (Total Addressable Market), SAM (Serviceable Available Market), and SOM (Serviceable Obtainable Market)

C) Revenue model

D) Technical roadmap from MVP to scaled architecture

Answer Key

Module 1 (Questions 1-10)
1. B
2. C
3. C
4. C
5. B
6. B
7. B
8. B
9. D
10. B

Module 2 (Questions 11-20)
11. C
12. C
13. C
14. B
15. B
16. B
17. C
18. B
19. C
20. B

Module 3 (Questions 21-30)
21. C
22. B
23. C
24. B
25. B
26. A
27. B
28. B
29. B
30. B

Synthesis (Questions 31-40)
31. C
32. B
33. D
34. C
35. B
36. C
37. B
38. D
39. B
40. B

Advanced Application (Questions 41-50)
41. A
42. B
43. B
44. D

45. B
46. B
47. B
48. B
49. D
50. B

Final Integration (Questions 51-60)
51. B
52. C
53. B
54. B
55. B
56. B
57. B
58. B
59. B
60. B

QUIZ SAMPLE 2